LESSON PLAN

PROPELLER THEORY (7)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Constant Speed Unit (CSU)
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Constant Speed Unit (CSU)
5. Specific Objectives	7.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of CSU cutaway diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller in an aircraft?
(b) How does a fixed pitch propeller differ from a variable pitch propeller?
(c) Why is a constant speed propeller preferred for modern aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of propellers • Fixed pitch vs Variable pitch • Constant speed unit components • Governor operation principles • Hydraulic pitch change mechanism	L & Ds	PPT, IB	15 min
(b)	• CSU internal valve operation • Oil pressure and pitch relationship • Low speed and high speed stops • Blade angle adjustment logic • RPM maintenance during flight	L & Ds	PPT, PDF	20 min
(c)	• Pitch change mechanisms • Hydraulic actuation systems • Feedback loops in CSU • Failure modes and indications • Maintenance considerations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the CSU maintains a constant RPM despite changes in engine power demand.
(b) Describe the role of oil pressure in changing the propeller blade pitch.
(c) What happens to the propeller pitch if the governor fails to maintain speed?

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of CSU operation apply to the counteracting methods discussed in LP 05 regarding propeller rotation effects.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Effect of propeller rotation on aircraft and methods used to counteract it	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min